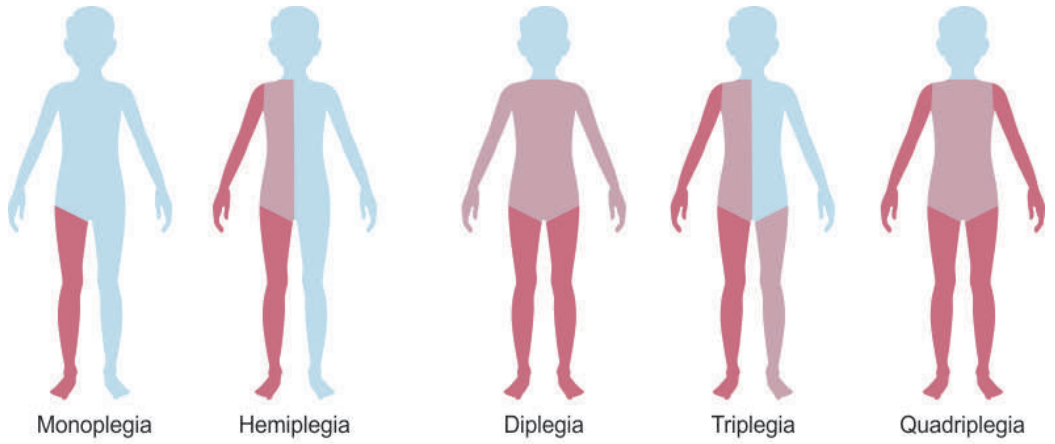
Early Signs

- ☑ Abnormal primitive reflexes
- ☑ Abnormal tone
- ☑ Persistent cortical thump
- ☑ Exaggerated startle reflex
- ☑ Delayed milestones
- ☑ Neonatal seizures
- ☑ Early handedness
- ☑ Feeding difficulties
- ☑ Sleep disturbance
- ☑ Inconsolable cry/easy irritability.

Classification

Physiological	*Topographical*
☑ Spastic	☑ Diplegia
☑ Dyskinetic	☑ Hemiplegia
– Dystonic	☑ Quadriplegia
– Choreoathetotic	☑ Monoplegia
☑ Ataxic	☑ Triplegia
☑ Mixed	☑ Double hemiplegia

Topographic Patterns of Cerebral Palsy

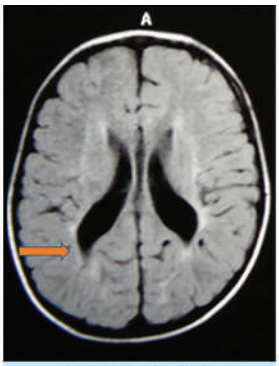


Clinical Features

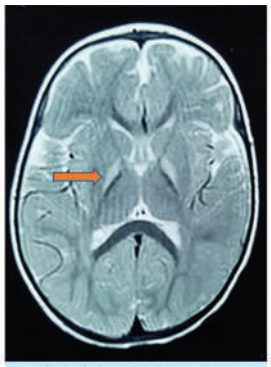
Neurological features	*Associated impairments/comorbidities*
Muscle weakness	Intellectual disability
Abnormal muscle tone	Epilepsy
Balance problems	Vision impairment
Loss of selective control	Hearing impairment
Pathological reflexes	Speech and communication problem
Loss of sensation	Swallowing difficulty
Contractures	Failure to thrive
Deformities	Respiratory problems
	Sleep disorders
	Oral dental caries/malocclusion
	Recurrent urinary tract infection and urinary incontinence
	Behavioral problems
	Anemia and malnutrition

Imaging Findings

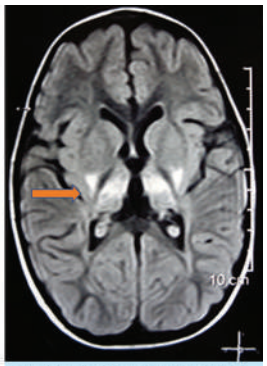
Focal encephalomalacia: Hemiplegic cerebral palsy



Periventricular leukomalacia: Spastic diplegic cerebral palsy



Kernicterus affecting globus pallidus: Dyskinetic cerebral palsy



Term asphyxia involving thalamus and posterior putamen: dyskinetic cerebral palsy

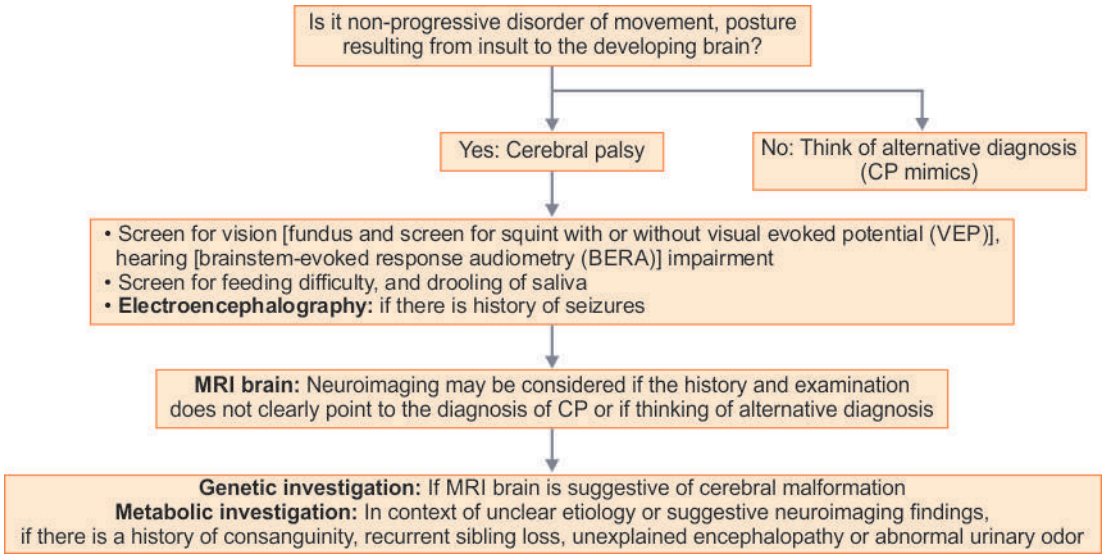
Common metabolic and genetic disorders that mimic CP according to the prominent motor dysfunction are as follows:

Disorders with prominent spasticity	***Disorders with prominent dyskinesia***	***Disorders with prominent ataxia***
☑ Hereditary spastic paraplegias ☑ Arginase deficiency ☑ COL4A1-related spastic CP ☑ Biotinidase deficiency ☑ Aicardi–Goutières syndrome ☑ Sulfite oxidase deficiency ☑ Molybdenum cofactor deficiency ☑ Leukodystrophies, such as metachromatic leukodystrophy ☑ Adrenoleukodystrophy ☑ Sjögren–Larsson syndrome	☑ Dopa-responsive dystonia ☑ Sepiapterin reductase deficiency ☑ Glutaric aciduria type 1 ☑ Glucose transporter deficiency type 1 ☑ Neurodegeneration with brain iron accumulation ☑ Cerebral creatine deficiency syndrome ☑ Lesch–Nyhan syndrome ☑ Cerebral folate deficiency ☑ ADCY5-related dyskinesia ☑ PCDH12-related dyskinesia ☑ NKX2-1-related ataxic dyskinetic CP ☑ TSEN54 gene-related pontocerebellar hypoplasia type 2	☑ Glucose transporter deficiency type 1 ☑ Ataxia telangiectasia ☑ Pelizaeus—Merzbacher disease ☑ Hereditary ataxias ☑ Joubert syndrome ☑ Mitochondrial cytopathies ☑ Pontocerebellar hypoplasia ☑ Cockayne syndrome ☑ Niemann—Pick disease type C ☑ Angelman syndrome ☑ Gangliosidosis type 1, juvenile and adult forms ☑ Nonketotic hyperglycinemia—Maple syrup urine disease NKX2-1-related ataxic dyskinetic CP

Red Flag Signs to Suspect Cerebral Palsy Mimickers

History	*Examination*
☑ No risk factors during perinatal period (such as prematurity, multiple gestation, low birth weight, multiple births, metabolic problems, kernicterus, intrapartum asphyxia, intracranial hemorrhage, infection, stroke, or head injuries) ☑ Positive family history of CP ☑ Fluctuation in motor symptoms ☑ Paroxysmal symptoms in relation to time of day, diet/fasting, or activity ☑ Progressive neurological symptoms ☑ Regression of milestones	☑ Dysmorphic features ☑ Isolated motor dysfunction ☑ Absent deep tendon reflexes ☑ Eye movement abnormalities (e.g., oculogyria, oculomotor apraxia, or paroxysmal saccadic eye-head movements) ☑ Optic atrophy ☑ Retinopathy

Diagnostic Algorithm for Evaluation of Cerebral Palsy



Management

Principle of Management

Management	*Components*
Physiotherapy	☑ It improves the postural control, improve muscle strength, range of motion (ROM) (decrease spasticity and contractures), joint alignment, and motor balance. ☑ Physiotherapy consists of exercise, bracing, and activities toward reaching specific functional goal. ☑ Physiotherapy consists of active and passive ROM exercises, stretching (prevent contractures), and strengthening (weak spastic and antagonist muscles)
Contracture management	☑ Dynamic contractures: casting, splinting, and use of botulinum toxin ☑ Fixed contractures: Tendon lengthening surgery
Botulinum toxin	☑ Botulinum toxin is used in patients who have increased muscle tone that interferes with function or is likely to lead to joint contracture with growth ☑ For patients with diffuse hypertonia, botulinum toxin can be injected into multiple muscles to treat the main foci of the generalized spasticity. ☑ Ideally, only two or three muscles will require treatment at one time. ☑ The patient should be re-evaluated 6–8 weeks after treatment. ☑ The effect typically lasts 3–8 months, after which treatment can be repeated.

Contd...

Principle of Management

Contd...

Management	Components
Intrathecal baclofen	For severely affected children who have significant side effects with antispasticity drugs or who have persistent and severe spasticity despite maximal doses, intrathecal baclofen may be an option and is an effective intervention. Side effects: Lethargy, confusion, hypotonia, and catheter-related complications
Medications for spasticity	☑ Baclofen (10–15 mg in three divided doses, maximum 40–60 mg) ☑ Tizanidine (2–4 mg/day in three to four divided doses, maximum of 35 mg/day) ☑ Diazepam (0.2–0.8 mg/kg/day in three to four divided doses)
Medications for dystonia	☑ Trihexyphenidyl (0.2–1 mg/kg/day) in three to four divided doses useful in dystonia. Other drugs that can be useful include gabapentin, baclofen, and diazepam. In refractory cases, deep brain stimulation (DBS) and pallidotomy may be useful in selected cases. ☑ Tetrabenazine (12.5–50 mg/day) for choreoathetoid movements
Drooling of saliva	Oral physiotherapy and use of ice cubes are most effective. In case of nonresponse, glycopyrrolate (0.5–1 mg/day), transdermal scopolamine, and botulinum toxin can be used.
Feeding difficulties and constipation	☑ Use of shallow spoon, use of frequent small feeds ☑ Oral physiotherapy for oromandibular dystonia ☑ Anti-gastroesophageal reflux disease (GERD) measures including metoclopramide and domperidone may be useful. In persistent feeding difficulties, options for percutaneous endoscopic gastrostomy (PEG) and feeding jejunostomy may be offered ☑ Constipation in children with cerebral palsy is often resistant to treatments. Polyethylene glycol high fiber diet may be useful for management of constipation.
Sleep and behavioral disturbances	☑ Good sleep hygiene must be explained to all parents. Medications including melatonin (3 mg), clonidine, and diphenhydramine may be useful in resistant cases ☑ Aggressive and disruptive behavior that impacts the quality of life may respond to risperidone (0.25–0.5 mg/day)
Bracing and orthosis	Ankle-foot orthosis (AFO) is useful among ambulatory children with cerebral palsy to improve their gait.
Surgery	☑ Selective dorsal rhizotomy is a surgical procedure that selectively divides parts of the dorsal lumbosacral roots of the spinal cord. It is useful among children with ambulatory children with spastic diplegia with good muscle strength with no deformities and contractures. ☑ Torsional deformities (femoral neck anteversion and tibial torsion) require rotational osteotomy usually done after 6 years of age. ☑ Hip dislocation might fail to respond to botulinum toxin or hip adductor release surgery.

Contd...

Principle of Management

Contd...

Management	Components
Deep brain stimulation	DBS is generally reserved for patients with severe dystonia that has not responded to medical therapy. DBS is achieved through a microelectrode that stimulates the internal segment of the globus pallidus
Advanced technology	☑ Augmentation technology where finger movements can drive the wheelchair ☑ Substitution technologies including speech technology ☑ Robotic-assisted gait trainers (RAGT) and robotic ankle training (RAT) can improve the gait and Gross Motor Function Classification System (GMFCS) in children with cerebral palsy ☑ Virtual reality including video games can be useful for motivating full body movement and has been found useful in balance problems and sensory perception problems ☑ Inertial navigation system to aid mobility that senses the motion of the subject
Occupational therapy	☑ Improve hand and upper extremity function in child through play and purposeful activity ☑ Improve self-care activities such as dressing, bathing, and brushing ☑ Play improves the mental capacity and provides psychological satisfaction ☑ Sports and recreation: Swimming and horseback riding (normalize muscle tone, decrease contractures, improve head control, and trunk balance)
Promising future treatment	Constraint-induced movement therapy (CIMT), robot-assisted activity, virtual reality, and functional electrical stimulation (FES). There is limited evidence on hippotherapy, hyperbaric oxygen therapy, systemic hypothermia, stem cell therapy, Adeli suit treatment, and acupuncture.

Management

Prognosis

Prognosis will depend on the type of CP (quadriplegic, diplegic, and hemiplegic), associated comorbidities (vision, hearing impairment, and epilepsy), developmental quotient, age at diagnosis, and the gross motor functional ability (GMFCS) and manual ability classification system (MACS). Children with GMFCS IV-V with presence of comorbidities will have a poor prognosis on motor outcome. Children who are able to achieve independent sitting by 2 years and independent walking by 4 years usually have better motor outcome.

Further Reading

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Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Breath-holding Spell

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